A New Algorithm for Land-Cover Classification Using PolSAR and InSAR Data and Its Application to Surface Roughness Mapping Along the Gulf Coast

Ke Wang, Jingyi Chen, *Member, IEEE*, Amin Kiaghadi, and Clint Dawson

***Abstract*— During a flooding event, the ability of the terrain to dissipate water flow energy depends on its land-cover type and the associated surface roughness. In this study, we developed a new land-cover classification algorithm using repeat-pass polarimetric synthetic aperture radar (PolSAR) and interferometric synthetic aperture radar (InSAR) data. Through a two-level hierarchical approach, we classified nine land-cover types with distinct surface roughness coefficients (Manning's n). We demonstrated the performance of this algorithm using available L-band ALOS PALSAR scenes acquired between April 2007 and April 2011 over the Houston area. The radar-based surface roughness estimates show a good agreement with those independently derived from NOAA's 22-class Coastal Change Analysis Program (C-CAP) 2010 land-cover classification data. Our algorithm is robust, and the randomly selected training sets only account for 0.3% of the total multilooked radar pixels (30-m spacing). Furthermore, we were able to accurately map surface roughness over the New Orleans area using available ALOS PALSAR scenes without selecting any new training sets. We note that NOAA's C-CAP data are currently used for estimating surface roughness in the operational storm-surge models, and a new version is typically released every five to six years. With the launch of the L-band NASA-ISRO Synthetic Aperture Radar (NISAR) mission in the near future, our algorithm can be used to fill the temporal gaps of the existing C-CAP-based surface roughness database and improve the accuracy of near real-time hydrodynamic modeling.**

***Index Terms*— Flood modeling, interferometric synthetic aperture radar (InSAR), land-cover classification, polarimetric synthetic aperture radar (PolSAR), surface roughness.**

I. Introduction

IN STORM surge modeling, surface roughness is a key parameter that influences drag forces exerted on the water flow by the land surfaces [1]–[3]. For example, a storm surge can travel much further inland over smooth surfaces, such as a barren landscape, than very rough surfaces, such as a dense forest. Because surface roughness depends on a wide range of surface properties, such as topographic conditions, the physical structure of obstructions, and vegetation density, it is not feasible to directly measure surface roughness over an entire flood plain. Instead, surface roughness data in storm surge models are derived from land-cover datasets [4], and large-scale surface roughness mapping can be considered as a land-cover classification problem.

Land-cover classification through field surveys is usually prohibitively expensive and time-consuming [5]. With the advantage of high-resolution and broad spatial coverage, satellite optical imagery has been used in urban scene classification, vegetation characterization, and surface water detection [6]–[8]. For example, NOAA's Coastal Change Analysis Program (C-CAP) land-cover data, known as the coastal expression of the National Land Cover Database (NLCD), were derived from Landsat imagery and ancillary *in situ* data [9]. The C-CAP data are currently used for estimating surface roughness in the Advanced Circulation (ADCIRC) modeling framework, an operational storm surge model for predicting overland flooding and inundation hours before the hurricane landfall [10]–[13]. A new version of C-CAP data is typically released every five to six years [9]. Because the quality and quantity of spaceborne imaging radar data have improved dramatically over the past decade, our goal is to develop a new land-cover classification algorithm for mapping surface roughness with improved temporal coverage using these data.

The existing radar-based land-cover classification algorithms have been mostly focused on the use of polarimetric synthetic aperture radar (PolSAR) amplitude data [14]–[22]. Spaceborne imaging data have not been used for estimating surface roughness in operational storm surge models because it is difficult to classify nine or more land cover types with distinct surface roughness with tens to hundreds of meters of spatial resolution over the entire model domain using the PolSAR amplitude data alone. In a recent study, Sica *et al.* [23] combined SAR amplitude and interferometric synthetic aperture radar (InSAR) temporal decorrelation characteristics to classify land surfaces into three groups (artificial surfaces, forests, and nonforest nature terrain). This is one of the few studies that include InSAR phase data as input for land-cover

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TABLE I
MANNING'S n VALUES FOR C-CAP LAND-COVER CLASSES [27]

Class Name	Manning's n [a]
Open Water	0.025
Bare Land	0.030
Unconsolidated Shore	0.030
Estuarine Aquatic Bed	0.030
Palustrine Aquatic Bed	0.035
Grassland/Herbaceous	0.035
Developed, Open Space	0.035
Estuarine Emergent Wetland	0.050
Pasture/Hay	0.055
Developed, Low Intensity	0.070
Palustrine Emergent Wetland (Persistent)	0.070
Estuarine Scrub/Shrub Wetland	0.070
Palustrine Scrub/Shrub Wetland	0.075
Scrub/Shrub	0.080
Developed, Medium Intensity	0.100
Cultivated Crops	0.100
Developed, High Intensity	0.120
Palustrine Forested Wetland	0.150
Estuarine Forested Wetland	0.150
Deciduous Forest	0.160
Mixed Forest	0.170
Evergreen Forest	0.180

[a] The unit of the original Manning's n values is $\mathrm{m}^{1/3} \cdot \mathrm{s}^{-1}$ in SI units. In practice, the governing equations in the ADCIRC model were non-dimensionalized to scale the problems, and thus the Manning's n coefficient can be considered as a dimensionless number.

classification algorithms. In this article, we present a new land-cover classification algorithm that, for the first time, allows us to identify the nine critical land-cover classes with distinct surface roughness coefficients (Manning's n) for the applications of storm surge modeling along the Gulf Coast. We were able to achieve the desired performance through a hierarchical classification approach and a careful design of input data (e.g., PolSAR amplitude, InSAR average coherence, and InSAR temporal decorrelation rate estimates) that discriminate between target land-cover classes.

II. METHODOLOGY

A. Overview

The ADCIRC modeling framework is a finite element shallow-water model widely used in storm surge simulations to solve for water flows associated with hurricanes and tropical storms [24], [25]. In this hydrodynamic framework, the bottom friction term is parameterized by a spatially distributed roughness coefficient (Manning's n) [26]. Manning's n values in the ADCIRC model are currently derived from the 22-class C-CAP land-cover classification data (2010) based on the empirical table from the Coastal Bend Bays and Estuaries Program (CBBEP) (see Table I) [27].

Because our goal is to map surface roughness using a radar-based land-cover classification algorithm, it is not necessary to distinguish land-cover types with very similar or identical Manning's n values. In addition, it is easier to classify fewer land-cover classes using the same input data.

TABLE II
MANNING'S n VALUES FOR RADAR-BASED LAND-COVER CLASSES

Class Name	Manning's n
Water	0.025
Open land: Barren land	0.030
Open land: Grassland/Herbaceous	0.035
Open land: Pasture/Emergent wetland (low veg. dens.)	0.055
Open land: Shrub/Emergent wetland (high veg. dens.)	0.075
Open land: Cultivated land	0.100
Suburban	0.100
Urban	0.120
Forest	0.150

This allows us to group 22 C-CAP land-cover types into nine umbrella classes without any substantial reduction in the accuracy of surface roughness estimation (see Table II). Through a hierarchical classification approach, we first divided the land surface into five classes with increasing roughness coefficients: water, open land, suburban, urban, and forest land-cover classes. We then refined the open land class into five subclasses of different vegetation densities (see Fig. 1). At each classification level, our algorithm requires ten or more SAR scenes acquired over the same region to characterize the statistical properties of the radar return echoes. As a result, this algorithm is designed to estimate the "average" land-cover types over the entire SAR acquisition period, and it cannot be used to capture any land-cover changes that occurred during this period. For example, if there are ten repeated-pass SAR images available every year, then this algorithm can be employed to derive annual land-cover maps. However, it cannot be used to track land-cover changes that occurred within a given year. Similarly, if there are ten repeated-pass SAR images available for each season of interest, this algorithm can be used to study seasonal land-cover and surface roughness changes.

B. Level-I Classification Algorithm

Sica *et al.* [23] have shown that InSAR temporal decorrelation characteristics, when combined with SAR amplitude measurements, can be used to classify land surfaces into artificial surfaces, forests, and nonforested nature terrain groups. Similarly, here we processed ten or more repeat-pass SAR images over an area of interest and computed the average SAR amplitude, average InSAR coherence, and InSAR temporal decorrelation rate estimates of all polarimetric channels as the input data to distinguish five land-cover classes with increasing roughness coefficients as follows.

An SAR image can be represented as a matrix S, where S_{ij} is a complex measurement at a pixel (i, j). If there are K repeat-pass SAR acquisitions available over the same area of interest, the average SAR amplitude at the pixel (i, j) is defined as

$$\bar{A}_{ij} = \frac{1}{K} \sum_{k=1}^{K} \left| S_{k,ij} \right| \tag{1}$$

where $S_{k,ij}$ is the SAR measurement at the pixel (i, j) from the kth SAR acquisition. While the observed SAR amplitude

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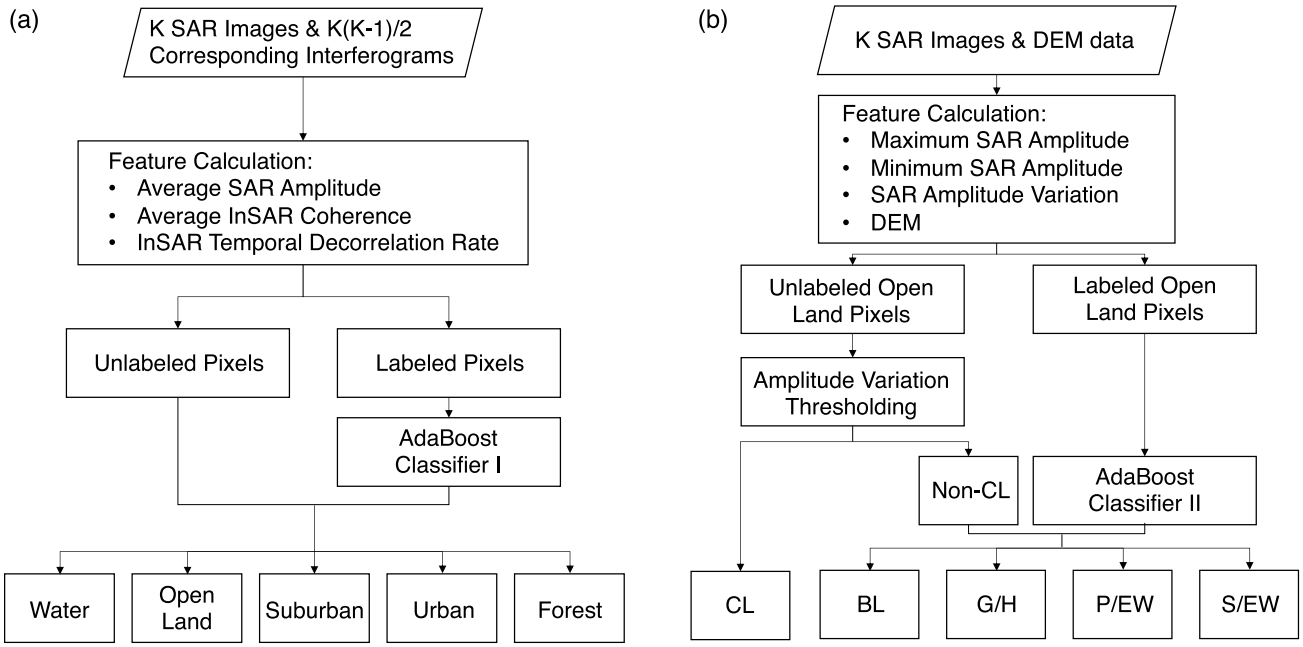

Fig. 1. Flow diagram of (a) level-I classification and (b) level-II classification. CL: cultivated land; BL: barren land; G/H: grassland/herbaceous; P/EW: pasture/emergent wetland (low vegetation density); and S/EW: shrub/emergent wetland (high vegetation density).

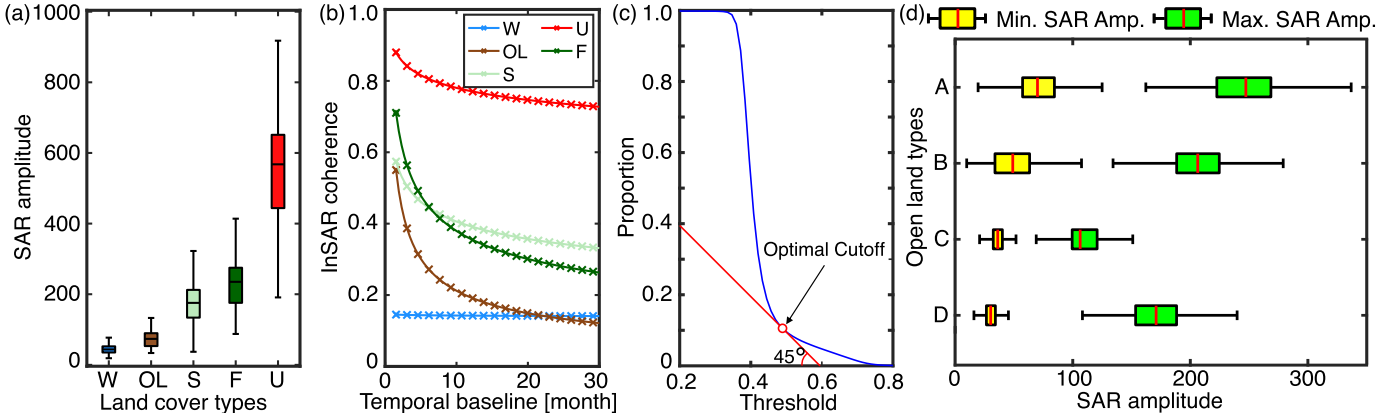

Fig. 2. (a) Box plot of typical L-band SAR amplitude measurements for the five targeted land covers. (b) Typical L-band coherence as a function of temporal baseline for the five targeted land covers. W: water; OL: open land; S: suburban; F: forest, and U: urban. (c) Proportion of pixels classified as cultivated lands as the SAR amplitude variation threshold varies between 0 and 1 (in blue). The threshold at which the tangent of the tradeoff curve equals 45° (in red) is defined as the amplitude variation cutoff for the classification of cultivated lands. (d) Typical range of L-band maximum (green boxes) and minimum (yellow boxes) SAR amplitude variation for four open land subclasses. A: shrub/emergent wetland (high vegetation density); B: pasture/emergent wetland (low vegetation density); C: grassland/herbaceous; and D: barren land.

at a given pixel of interest can vary significantly from multiple SAR acquisitions, the average SAR amplitude typically correlates with the land-cover class of that pixel [see Fig. 2(a)]. When PolSAR data are available, the average SAR amplitude of each polarization channel can be used as independent input data to further improve the separability of land-cover classes. In general, horizontal–vertical (HV) polarization is more effective at distinguishing forest/nonforest land covers [28]–[30], and horizontal–horizontal (HH) polarization is more effective at distinguishing water/land covers [31], [32].

We can form a maximum of $K \times (K-1)/2$ interferograms using K SAR images acquired over the same region with the same look angle. Given an interferogram that is formed using two SAR images S_1 acquired at the time t_1 and S_2 acquired at the time t_2, the corresponding InSAR coherence at the pixel (i, j) can be computed as [33]

$$\hat{C}_{12,ij} = \frac{\left|\sum_{(p,q)\in b(i,j)} S_{1,pq} S_{2,pq}^{*}\right|}{\sqrt{\sum_{(p,q)\in b(i,j)} |S_{1,pq}|^2} \sqrt{\sum_{(p,q)\in b(i,j)} |S_{2,pq}|^2}} \quad (2)$$

where

$$b(i,j) = \{(p,q) | i - w \le p \le i + w \\ j - w \le q \le j + w\} \quad (3)$$

defines an image block centered at the pixel (i, j) with a half width w.

In general, the observed InSAR coherence can be written as a product of thermal, spatial, and temporal coherences [34]. Because the signal-to-noise ratio (SNR) of the space-borne InSAR data is typically high, thermal coherence is

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typically ≈ 1. Spatial coherence decreases as the perpendicular component of the spatial baseline B_{perp} increases, and temporal coherence decreases as the temporal baseline increases. At the pixel (i, j), we define the average InSAR coherence $\bar{C}_{ij}$ of all interferograms as

$$\bar{C}_{ij} = \frac{1}{L}\sum_{l=1}^{L} \bar{C}_{ij}(\delta t_l) \tag{4}$$

where $\bar{C}_{ij}(\delta t_l)$ is the average InSAR coherence of a subset of interferograms with the same temporal baseline δt_l. L is the number of unique δt_l values among all $K \times (K-1)/2$ interferograms. Note that there are different numbers of interferograms in each of the L interferogram subsets, and (4) assigns each subset an equal weight of $1/L$. Typically, the average InSAR coherence increases in the order of water, open land, forest, suburban, and urban classes [see Fig. 2(b)].

We estimated the optimal InSAR temporal decorrelation rate $\hat{\alpha}_{ij}$ and a scale factor $\hat{\beta}_{ij}$ at the pixel pixel (i, j) by fitting the InSAR coherence observations to a power-law decay model (see the Appendix) as

$$(\hat{\alpha}_{ij}, \hat{\beta}_{ij}) = \underset{\alpha_{ij},\beta_{ij}}{\arg\min} \sum_{l=1}^{L} \left(\bar{C}_{ij}(\delta t_l) - \beta_{ij}\delta t_l^{-\alpha_{ij}}\right)^2. \tag{5}$$

Here, the coherence observations of interferograms with the same temporal baseline are averaged ($\bar{C}_{ij}(\delta t_l)$) to reduce the uncertainty in $\hat{\alpha}_{ij}$ introduced by other decorrelation factors, such as spatial decorrelation. In general, the estimated InSAR temporal decorrelation rate $\hat{\alpha}_{ij}$ increases in the order of water, urban, suburban, forest, and open land classes [see Fig. 2(b)]. Because both the scale factor $\hat{\beta}_{ij}$ and the average InSAR coherence $\bar{C}_{ij}$ characterize the magnitude of the InSAR coherence, $\hat{\beta}_{ij}$ is not used as input data for the land-cover classification to avoid redundant input feature. When PolSAR data are available, the average InSAR coherence and InSAR temporal decorrelation rate estimates from each polarimetric channel can be used as independent input data for the classification algorithm.

The level-I classification is performed at each pixel location using the decision-tree-based adaptive boosting (AdaBoost) algorithm [35], a supervised learning algorithm that requires a set of labeled training pixels. This training set can be randomly selected based on existing land-cover data or optical imagery (e.g., Landsat data) acquired during the study period. For each training pixel with a known land-cover label, the average SAR amplitude, the average InSAR coherence, and the InSAR temporal decorrelation rate estimates from available polarimetric channels are used as its feature vector to train the AdaBoost classifier. After the training phase, the feature vector of an unclassified pixel is used to determine its land-cover class. The AdaBoost algorithm has the advantage of combining multiple weak classifiers into an accurate strong classifier [36]. It is also known to be less susceptible to the overfitting problem than classification algorithms, such as the support vector machine (SVM) and bagging algorithms [37], [38]. This property allows us to reduce the number of required training pixels and the complexity of the training process.

C. Level-II Classification Algorithm

The open land class contains five subclasses with increasing Manning's n values due to the differences in the vegetation density: barren land, grassland/herbaceous, pasture/emergent wetland (low vegetation density), shrub/emergent wetland (high vegetation density), and cultivated land (see Table II). Because they have similar average SAR amplitude, average InSAR coherence, and InSAR temporal decorrelation rate measurements from all polarimetric channels, these land-cover classes cannot be separated in the Level-I classification. Here, we designed the Level-II classification algorithm to classify these open land subclasses.

Cultivated lands typically have larger SAR amplitude variation (due to the growth and harvest of crops) than other open land types. Given a set of SAR images $S_1, \ldots, S_K$, the SAR amplitude variation is defined as

$$V_{ij} = \frac{A_{\max,ij} - A_{\min,ij}}{A_{\max,ij} + A_{\min,ij}} \tag{6}$$

where $A_{\max,ij}$ and $A_{\min,ij}$ are the maximum and minimum of all K SAR amplitude measurements $|S_{1,ij}|, \ldots, |S_{K,ij}|$, respectively. A radar pixel (i, j) is classified as cultivated land if its amplitude variation V_{ij} is greater than a fixed threshold. This threshold is chosen from a range of candidate thresholds between 0 and 1 as follows. As the threshold increases, the ratio of the number of cultivated land pixels to the total number of pixels decreases [see Fig. 2(c)]. The threshold value at which the tangent of the tradeoff curve equals 45° is chosen as the amplitude variation cutoff for the classification of cultivated lands. In the case that multiple polarimetric channels are available, a radar pixel is marked as cultivated land if its amplitude variation is greater than the cutoff threshold in any of the polarimetric channels. Note that some coastal emergent wetlands and barren lands also show high amplitude variation due to tides. We excluded these low-lying areas below an elevation threshold of 3 m using the Shuttle Radar Topography Mission digital elevation data (SRTM DEM) [39].

The remaining open land pixels are grouped into four subclasses of different vegetation densities. Given that these classes have similar coherence characteristics, only the maximum and minimum SAR amplitude and the SAR amplitude variation are used as input data in the level-II classification. While barren lands and grasslands both have low minimum SAR amplitude, barren lands typically have larger amplitude variation and higher maximum amplitude. For areas with higher vegetation density (e.g., shrubs), both minimum and maximum amplitudes are typically higher [see Fig. 2(d)]. Again, the SAR amplitude measurements from multiple polarization channels can be used as separate input data for the classification. In addition to information derived from PolSAR amplitude measurements, SRTM DEM data are used as an auxiliary classifier to distinguish land-cover types concentrated on areas with low elevation, such as emergent wetlands. The AdaBoost classification algorithm is performed on all open land pixels detected in level-I classification (excluding the cultivated land class) to complete level-II classification.

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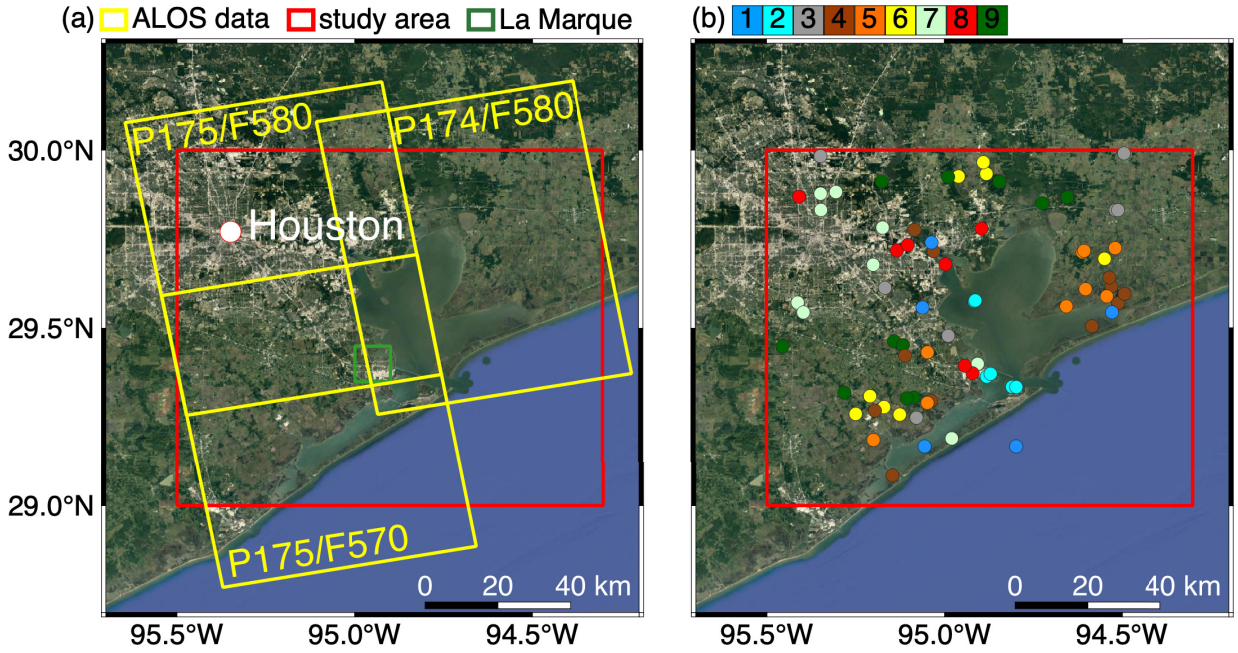

Fig. 3. (a) Google Earth image of the Houston area. The area of interest is outlined in red, and the ALOS PALSAR spatial coverage over the study site is outlined in yellow. ALOS path and frame numbers are marked on each radar scene. The La Marque area outlined in green is magnified in Fig. 4 for the illustration of the average SAR amplitude, the average InSAR coherence, and InSAR temporal decorrelation rate estimates. (b) Location of training polygons used for classification. Different land-cover types are distinguished by colors. 1: water; 2: barren land; 3: grassland/herbaceous; 4: pasture/emergent wetland (low vegetation density); 5: shrub/emergent wetland (high vegetation density); 6: cultivated land; 7: suburban; 8: urban; and 9: forest.

III. Test Sites and Available Data

We tested our radar-based nine-class land-cover classification algorithm over Houston, TX, USA. The study area was hard-hit by recent flooding events, and accurate and up-to-date surface roughness data are critical for future flood risk analysis and near-real-time decision support. We processed all available L-band ALOS PALSAR data (12 scenes from Path 175 Frame 570–580 and 13 scenes from Path 174 Frame 580) acquired between April 2007 and April 2011 over the area of interest [see Fig. 3(a)]. We calculated the average SAR amplitude for each polarization (HH/HV) and acquisition mode (FBS/FBD), and the average InSAR coherence for each polarization (HH/HV). Due to the limited availability of HV-polarized interferograms, we were only able to reliably estimate the InSAR temporal decorrelation rate of the HH-polarized interferograms. These radar-based measurements (e.g., Fig. 4) were resampled to a 30-m latitude/longitude grid, and they were used as the input data in the level-I classification. We further performed the level-II classification on all open land pixels, in which the maximum and minimum SAR amplitude and the SAR amplitude variation for each acquisition mode and polarization and the SRTM DEM were used as input data. The input features used in the level-I and level-II classification are summarized in Table III.

To train the level-I and level-II classifiers, we labeled five or more polygons for each land-cover class in Table II using high-resolution Landsat images acquired around 2010. Multiple Landsat images were inspected to avoid selecting areas with substantial land-cover changes. Because open land subclasses may appear similar in Landsat images, C-CAP 2010 land-cover data were also used to select training samples in the level-II classification. To ensure each polygon only contains pixels from a single land-cover class, we did not select any training pixels from transitional areas that may contain multiple land-cover classes. In summary, 74 training polygons that contain 72 331 pixels were selected as the training set, and 29 861 436 pixels were classified into the nine land-cover classes.

Based on Table II, we assigned each land-cover class a Manning's n value, and we projected the roughness map on the ADCIRC finite element mesh nodes. We also derived surface roughness maps from the 22-class C-CAP 2010 land-cover data using Table I and projected the roughness map on the same mesh to validate our algorithm. Both C-CAP 2010 and ALOS surface roughness estimates were derived using data acquired around 2010, and they do not capture changes that occurred within a time scale of five years or less. This is because: 1) the ALOS-derived surface roughness estimates are considered as the "average" surface roughness between April 2007 and April 2011 and 2) C-CAP 2010 data were derived using Landsat and *in situ* data collected around 2010, and a new C-CAP version is only released every five to six years.

IV. Results

Overall, the nine-class ALOS PALSAR land-cover maps [see Fig. 5(a)] and the 22-class C-CAP land-cover maps (2010) yield consistent surface roughens estimates [see Fig. 5(b) and (c)] at the spatial scale of 5 km or more. Pearson's correlation coefficient R [40] between the ALOS-derived and C-CAP-derived surface roughness estimates is 0.90, with p value < 0.01. Here, Pearson's coefficient R ranges from -1 to 1, where 1 means a perfect linear relationship, 0 means no correlation, and -1 means a perfect

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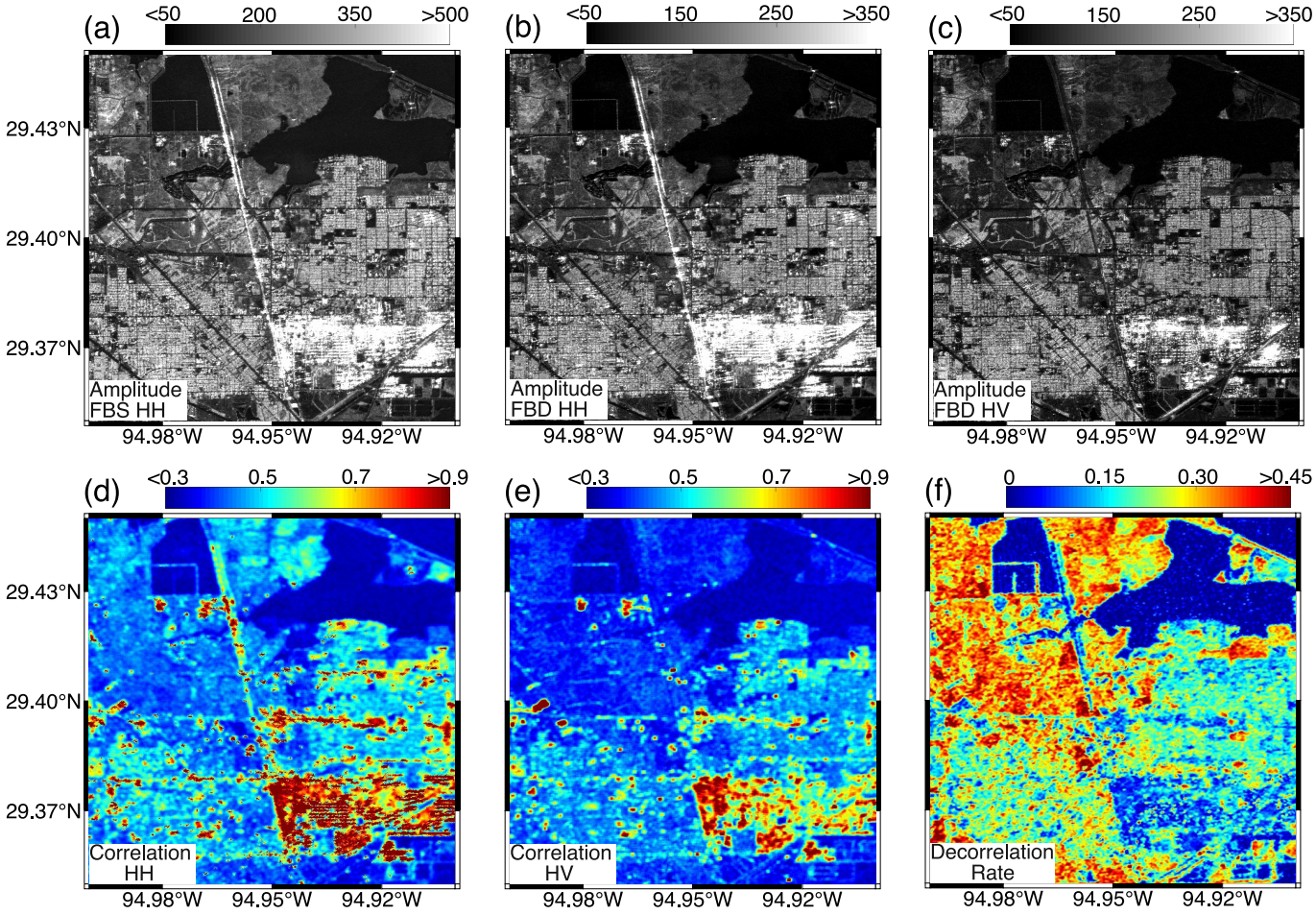

Fig. 4. (a)–(c) Average SAR amplitude of all ALOS PALSAR FBS-mode HH-polarized scenes, FBD-mode HH-polarized scenes, and FBD-mode HV-polarized scenes over the La Marque area [outlined in green in Fig. 3(a)]. (d) and (e) Average InSAR coherence of HH-polarized interferograms, and HV-polarized interferograms. (f) Estimated InSAR temporal decorrelation rate of HH-polarized interferograms.

TABLE III
INPUT FEATURES USED FOR LEVEL-I AND LEVEL-II CLASSIFICATIONS

Level	Feature	Feature Name	Available Polarization
Level-I	1-3	Average SAR amplitude	FBS-HH, FBD-HH, FBD-HV
	4-5	Average InSAR coherence	HH, HV
	6	InSAR temporal decorrelation rate	HH
Level-II	7-9	Maximum SAR amplitude	FBS-HH, FBD-HH, FBD-HV
	10-12	Minimum SAR amplitude	FBS-HH, FBD-HH, FBD-HV
	13-15	SAR amplitude variation	FBS-HH, FBD-HH, FBD-HV
	16	SRTM DEM	-

inverse linear relationship. The p value represents the significance of the correlation between the two surface roughness estimates. The correlation is usually considered statistically significant if p value ≤ 0.05 [41].

The differences between the two estimates [see Fig. 5(d)] are mainly due to the disagreement on cultivated land classification between the two land-cover datasets at spatial scales of 1 km or less. The classification discrepancies of cultivated lands at this spatial scale are common in the existing remote sensing land-cover database. This is because surface properties of cultivated lands often change substantially at very fine spatial scales over an agricultural cycle. As an example, Fig. 6 shows the identified cultivated land pixels from the ALOS, C-CAP 2010, and NLCD 2011 classifications over the same region. Using a 2010 Landsat image as ground truth, we found that both ALOS and NLCD 2011 data identified the flooded field as open water with a very low Manning's n value (0.025), while C-CAP 2010 data incorrectly identified these pixels as cultivated lands with a much higher roughness coefficient (0.1). ALOS data also identified fallow fields as barren lands or grassland instead of cultivated lands. Because the level-II classification is designed to separate open land

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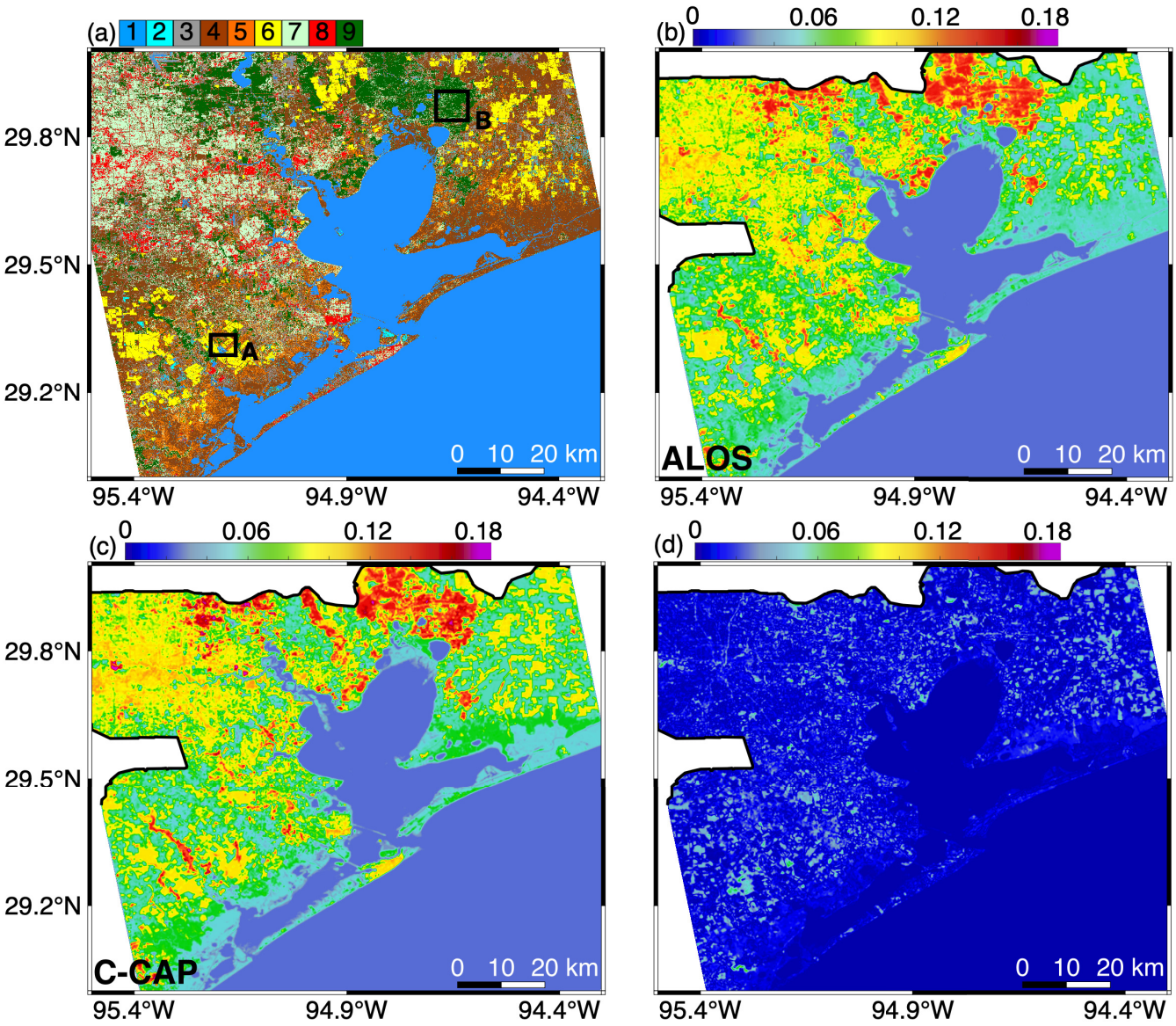

Fig. 5. (a) Radar-based land-cover classification results over the Houston area. The two patches outlined in black are magnified in Figs. 6 and 7. Different land-cover types are distinguished by colors. 1: water; 2: barren land; 3: grassland/herbaceous; 4: pasture/emergent wetland (low vegetation density); 5: shrub/emergent wetland (high vegetation density); 6: cultivated land; 7: suburban; 8: urban; and 9: forest. Manning's n values projected on ADCIRC finite element mesh nodes as derived from (b) ALOS PALSAR data and (c) NOAA's C-CAP 2010 land-cover classification data. (d) Absolute difference between ALOS- and C-CAP derived Manning's n. The boundary of the computational domain is outlined in black in (b)–(d).

pixels with different vegetation densities, our algorithm can accurately identify agricultural fields with different surface roughness coefficients, whereas C-CAP or NLCD data tend to classify agricultural fields as cultivated lands and overestimate the surface roughness for flooded or fallow fields.

The differences between the C-CAP and ALOS surface roughness estimates can be also introduced in the process of merging the 22 C-CAP land-cover types into the nine umbrella classes. For example, the forest class in the ALOS classification can be further divided into four subclasses with increasing Manning's n values (e.g., Fig. 7): estuarine/palustrine forested wetland (0.15), deciduous forest (0.16), mixed forest (0.17), and evergreen forest (0.18) (see Table I). Using the L-band ALOS data alone, we can identify forest pixels and nonforest pixels, but we cannot distinguish the foliage differences. Because the Houston area is located in the low-lying coastal regions with a humid subtropical climate, the majority of the forest pixels are forest wetlands. Therefore, we assigned 0.15 as Manning's n value for the forest class in this study. Given that radar signals of different wavelengths and multiple polarimetric channels have different penetration depths and scattering mechanisms, multipolarimetric and multifrequency SAR and InSAR measurements, if available, can be used to construct the feature vector to improve the classification of forest subclasses [42]–[45]. For example, forests with denser canopy, such as evergreen forests, tend to have a substantial volume scattering component in the radar return signals. Forested wetlands, when they are flooded, are usually dominated by double-bounce scattering due to the water/trunk interfaces.

While multipolarimetric and multifrequency SAR and InSAR measurements can be used as additional input features

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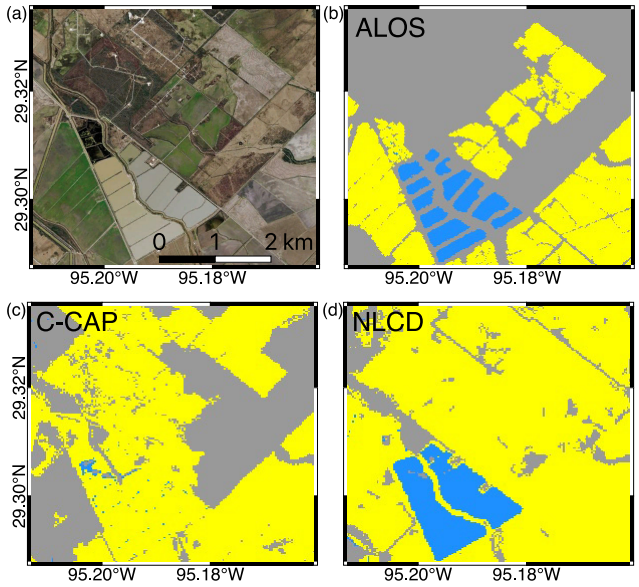

Fig. 6. Comparison of cultivated land classification. (a) Landsat image acquired in January 2010 over the area outlined in black (patch A) in Fig. 5(a); (b) ALOS, (c) C-CAP 2010, and (d) NLCD 2011 land-cover classification map over the same region. Land-cover types are marked in different colors: blue: water; yellow: cultivated land; and gray: noncultivated land.

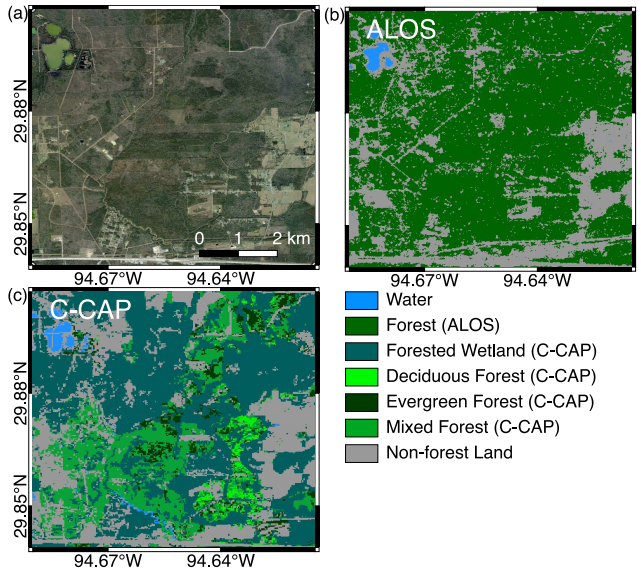

Fig. 7. Comparison of forest classification between ALOS and C-CAP 2010 land-cover classification data. (a) Landsat image acquired in January 2010 over the area outlined in black (patch B) in Fig. 5(a); (b) ALOS and (c) C-CAP 2010 land-cover classification map over the same region.

to improve the classification of forest subclasses, we found that L-band data (e.g., ALOS PALSAR) produce more accurate surface roughness estimates than C-band or X-band data (e.g., Sentinel-1 or TerraSAR-X) when used alone. This is mainly because C-band or X-band InSAR phase measurements decorrelate much faster than L-band InSAR phase measurements. As a result, interferograms with short wavelengths (X-band or C-band) have similar low average coherence and fast temporal decorrelation rate for open land, suburban, and forest classes. A robust classification algorithm requires input features that can discriminate between target land-cover classes.

V. Discussion

A. Role of Training Sets

Our algorithm does not require extensive human efforts in the selection of training samples. For the Houston case, a total of 74 polygons that contain 72 331 pixels were selected as the training set, which only accounts for 0.3% of the total number of classified pixels (see Table IV). To test the robustness of our algorithm, we also analyzed all available L-band ALOS PALSAR data (19 scenes from Path 165 Frame 590 and 19 scenes from Path 166 Frame 580) acquired between April 2007 and April 2011 over the New Orleans area [see Fig. 8(a)]. As in the Houston case, we calculated the feature vectors for the level-I and level-II classification at each 30×30 m^2 pixel (see Table III). An important difference here is that we did not select any new training pixels from the New Orleans area, and we classified 5 985 662 unlabeled New Orleans pixels into the nine land-cover classes using the classifiers that were trained by the Houston training set [see Fig. 8(b)]. The ALOS-based surface roughness estimates yield comparable results as the C-CAP estimates [see Fig. 8(c)–(e)]. Pearson's correlation coefficient between the two estimates is 0.92, with p value < 0.01. This indicates that our algorithm is robust, and our classification results do not depend on the random selection of training sets. We note that both Houston and New Orleans are located along the Gulf Coast with similar climate patterns and land-cover types. Additional training samples may be required if we apply this classification algorithm to other coastal areas with drastically different climate patterns and land-cover types.

B. Comparison Between the Two- and One-Level Classification Strategies

In this article, a two-level algorithm was developed to classify the radar pixels into nine land-cover classes. In the level-I classification, the average SAR amplitude, average InSAR coherence, and InSAR temporal decorrelation rate (Features 1–6 in Table III) were used as input data to classify the land surfaces into the water, open land, suburban, urban, and forest classes. In the level-II classification, we identified cultivated land pixels using an amplitude variation threshold, and then, we classified the remaining four subclasses with different vegetation densities based on the maximum and minimum SAR amplitude, the SAR amplitude variation, and the SRTM DEM data (Features 7–16 in Table III).

As a comparison, we also implemented a one-level classification strategy. In this algorithm, all 16 features listed in Table III were used as input data to classify radar pixels into eight land-cover classes: water, suburban, urban, forest, and four open land subclasses (barren land, grassland/herbaceous, pasture/emergent wetland, and shrub/emergent wetland). To derive the complete nine-class land-cover classification results, we employed the amplitude variation threshold method, as described in Section II-C, and identified the cultivated land pixels from the four open land subclasses.

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TABLE IV
TOTAL NUMBERS OF TRAINING AND CLASSIFIED PIXELS FOR NINE LAND-COVER CLASSES

Class Name	# of training pixels	# of classified pixels	
		Houston	New Orleans
Water	24,460	8,901,000	2,114,272
Barren land	3,499	339,665	14,480
Grassland/Herbaceous	4,185	1,503,426	70,035
Pasture/Emergent wetland (low veg. density)	17,508	8,281,552	1,031,333
Shrub/Emergent wetland (high veg. density)	2,480	1,496,538	846,224
Cultivated land	2,950	1,534,009	232,123
Suburban	8,182	3,704,393	531,883
Urban	1,447	1,014,485	130,203
Forest	7,600	3,086,368	1,015,109
Total	72,331	29,861,436	5,985,662

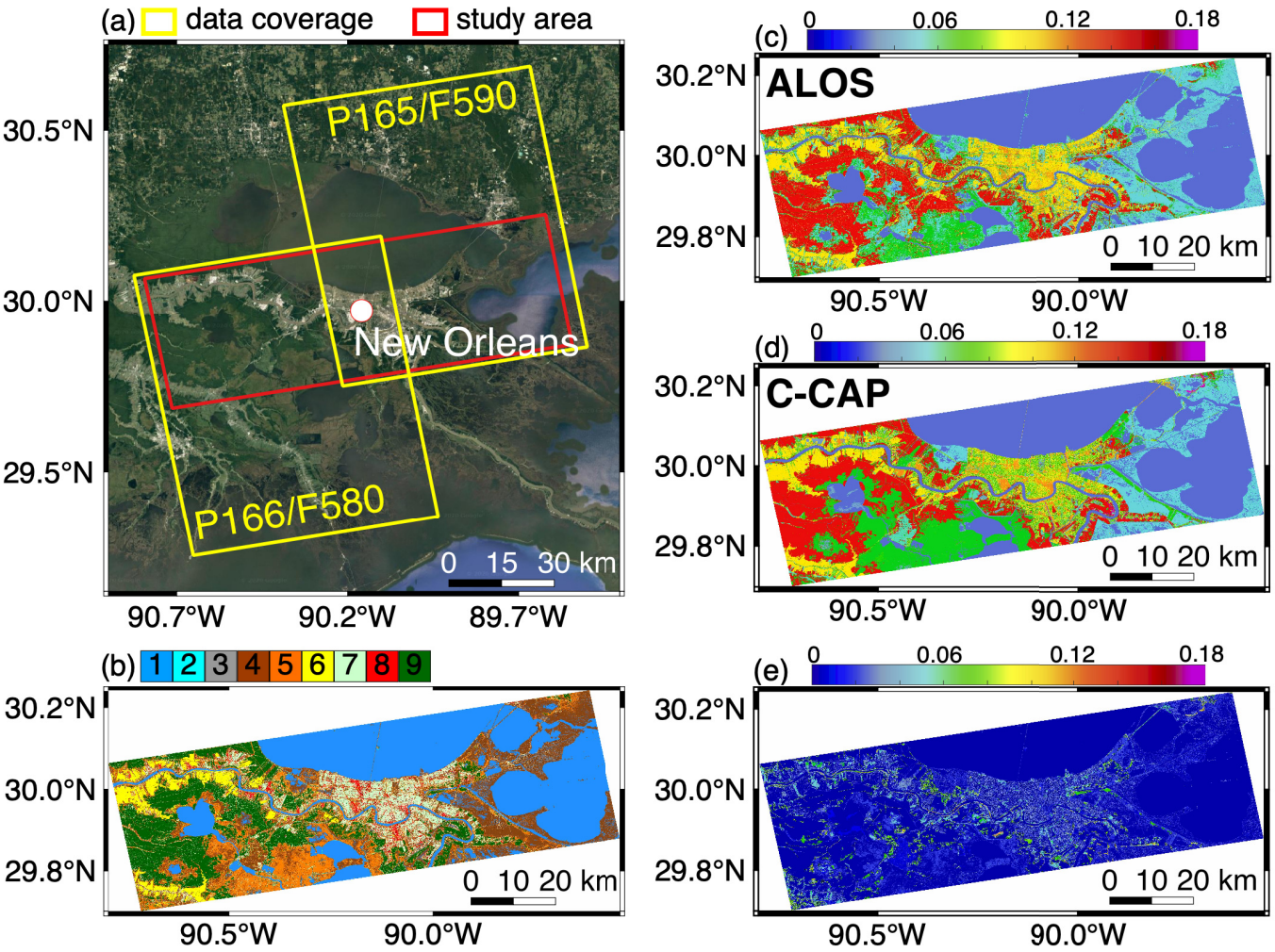

Fig. 8. (a) Google Earth image of the New Orleans area. The area of interest is outlined in red, and the ALOS PALSAR spatial coverage over the study site is outlined in yellow. ALOS path and frame numbers are marked on each radar scene. (b) Radar-based land-cover classification results over the New Orleans area. Different land-cover types are distinguished by colors. 1: water; 2: barren land; 3: grassland/herbaceous; 4: pasture/emergent wetland (low vegetation density); 5: shrub/emergent wetland (high vegetation density); 6: cultivated land; 7: suburban; 8: urban; and 9: forest. (c) Manning's n map derived from ALOS PALSAR data. (d) Manning's n map derived from NOAA's C-CAP 2010 land-cover classification data. Warm color represents high Manning's n values. (e) Absolute difference between the ALOS and the C-CAP Manning's n estimates.

We applied the two- and one-level algorithms to analyze the ALOS Houston data [see Fig. 3 (a)]. In both cases, the same training data were used in the training phase [see Fig. 3(b)], and the land-cover classification results were translated to surface roughness estimates based on Table II. Pearson's correlation coefficient between the ALOS and C-CAP 2010 surface roughness estimates is 0.80 and 0.90 for the one- and two-level cases, respectively. As an example, Fig. 9 shows

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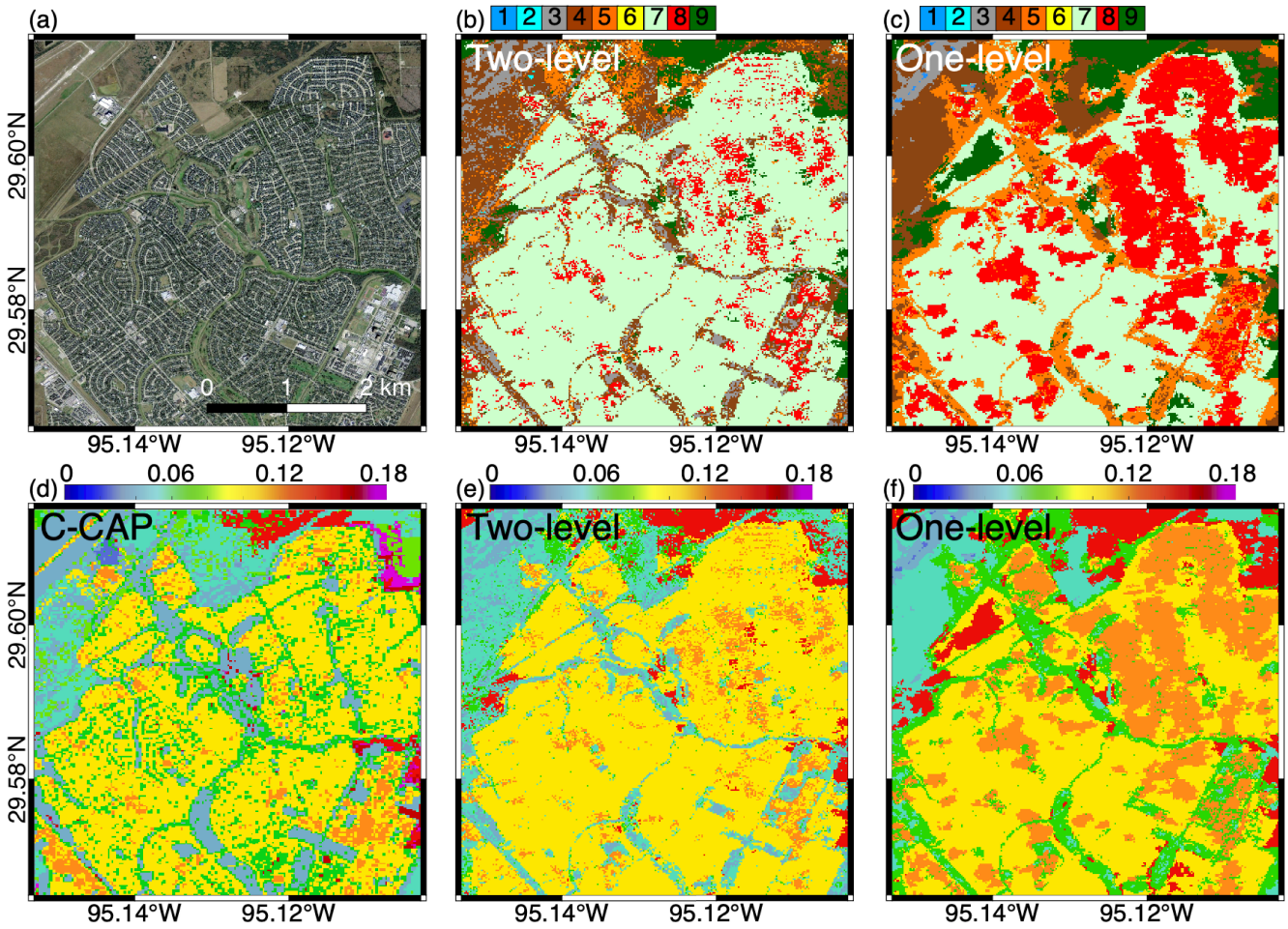

Fig. 9. (a) Google Earth image acquired in March 2010 over a focused study area in Clear Lake City, Houston. (b) Land-cover classification results derived from the two-level classification strategy. (c) Land-cover classification results derived from the one-level classification strategy. Land-cover types in (b) and (c) are marked by colors. 1: water; 2: barren land; 3: grassland/herbaceous; 4: pasture/emergent wetland (low vegetation density); 5: shrub/emergent wetland (high vegetation density); 6: cultivated land; 7: suburban; 8: urban; and 9: forest. Manning's n map derived from (d) C-CAP 2010 land-cover classification data, (e) two-level ALOS classification results, and (f) one-level ALOS classification results.

the land-cover classification results over an area in Clear Lake City, Houston. We found that the two-level classification algorithm produces more accurate surface roughness estimates than the one-level classification algorithm. One reason that the two-level classification outperforms one-level classification is that the two-level classification allows us to make use of the natural hierarchy of land-cover classes. For example, the five subclasses considered in the level-II classification not only have similar surface roughness but also are commonly characterized by areas covered with short vegetation. This natural similarity between these five subclasses enables us to group them into the open land class in the level-I classification. Ignoring this superclass/subclass relationship may reduce classification performance. For this reason, such multilevel classification approaches have been widely used for radar image classification applications, including wetland classification, ship classification, and general land-cover classification [46]–[48].

C. Impact of Temporal Decorrelation Models on Surface Roughness Estimates

As we stated in Section II-B, the observed InSAR phase coherence can be reduced due to thermal decorrelation, spatial decorrelation, and temporal decorrelation [34]. In addition, processing errors (e.g., errors in the pixel coregistration step) may also lead to decorrelation artifacts. Among those decorrelation factors, temporal decorrelation is the only one that is associated with the changes in the Earth's surface properties, which typically decreases as the temporal baseline increases. To reduce the influence of other decorrelation factors in the final temporal decorrelation rate estimates, we first averaged the observed InSAR phase coherence of the interferograms with the same temporal baseline. We then estimated the InSAR temporal decorrelation rate by fitting InSAR coherence observations to a power-law decay model [see (5)]

$$C_{\mathrm{Pow}}(\delta t) = \beta \delta t^{-\alpha} \tag{7}$$

where α is the InSAR temporal decorrelation rate and β is a scale factor.

In this section, our goal is to evaluate the impact of different temporal decorrelation models on surface roughness estimation accuracy. In particular, we considered two exponential temporal decorrelation models (the exponential model A and the exponential model B) for estimating the InSAR temporal

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TABLE V
COMPARISON BETWEEN DIFFERENT DECORRELATION MODELS

Model	Mean MSE	# of model parameters	Pearson's Correlation R	Computation Time
Pow[a]	0.0062	2	0.902	10.0 seconds
Pow[b]	0.0051	2	0.902	9.6 hours
ExpA	0.0058	2	0.896	11.5 hours
ExpB	0.0048	3	0.899	14.1 hours

[a] Power-law decay model with linear approximation
[b] Power-law decay model without linear approximation

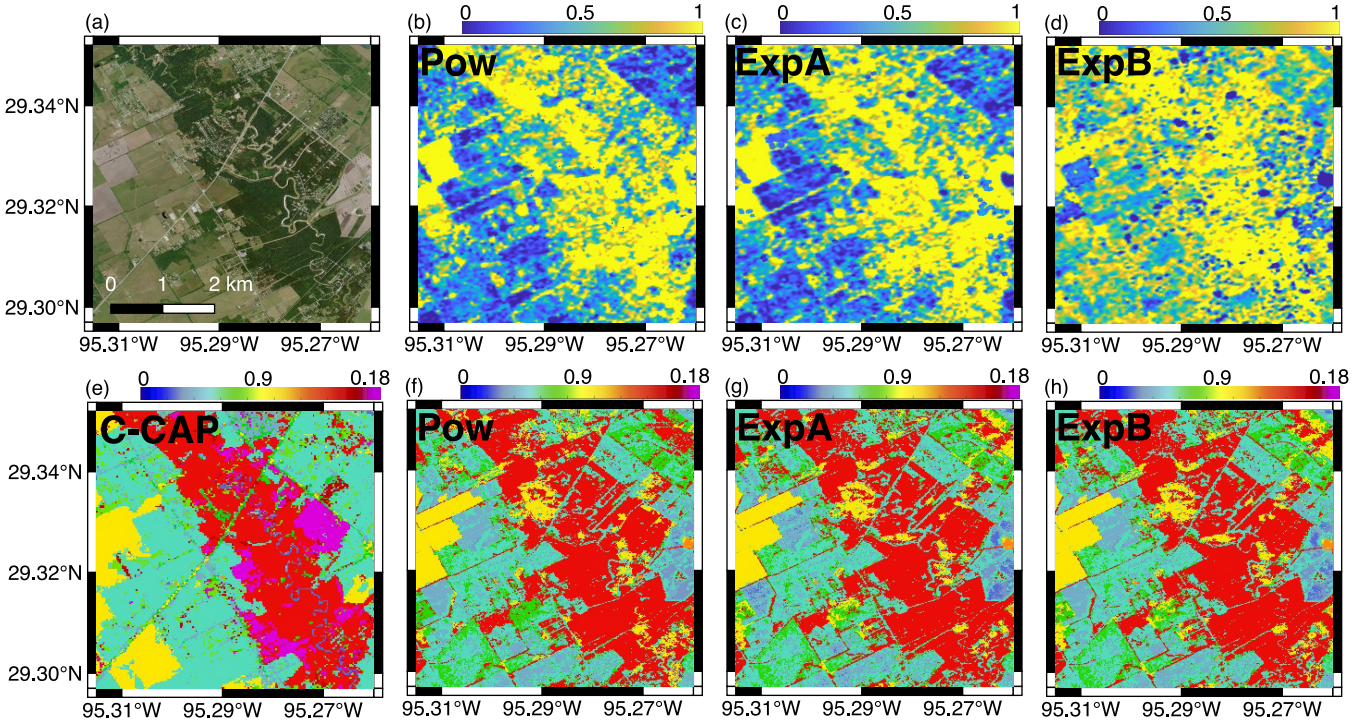

Fig. 10. (a) Google Earth image acquired in January 2010 over a focused study area. InSAR temporal decorrelation rate estimated using (b) power-law decay model, (c) exponential model A, and (d) exponential model B. (e) Manning's n map derived from the C-CAP 2010 land-cover classification data. Manning's n maps derived from the ALOS PALSAR data using the following temporal decorrelation models: (f) power-law decay model, (g) exponential model A, and (h) exponential model B.

decorrelation rate [23], [49], [50]

$$C_{\mathrm{ExpA}}(\delta t) = C_0\, e^{-\frac{\delta t}{\tau}} \tag{8}$$

$$C_{\mathrm{ExpB}}(\delta t) = (C_0 - C_{lt})e^{-\frac{\delta t}{\tau}} + C_{lt} \tag{9}$$

where C_0 represents the initial coherence and $1/\tau$ is the InSAR temporal decorrelation rate. The long-term coherence term C_{lt} is added in (9) to take into account the fact that radar pixels may not completely decorrelate even after a very long time.

We solved for the optimal model parameters of the three models in the least-squares sense using the nonlinear curve fitting MATLAB function `lsqcurvfit`. As shown in Table V, fitting the exponential models is more computationally expensive than fitting the power-law decay model. Furthermore, the nonlinear fitting problem of the power-law decay model can be approximated to a weighted linear regression problem (see the Appendix). This allows us to drastically reduce the computation time for estimating the InSAR temporal decorrelation rates of the Houston dataset (29 861 436 pixels) from 9.55 h to 10 s on the same workstation.

We calculated the mean square error (MSE) between InSAR coherence observations and model predictions at a pixel (i, j) as

$$\mathrm{MSE}_{M,ij} = \frac{1}{L}\sum_{l=1}^{L}\left(\bar{C}_{ij}(\delta t_l) - C_{M,ij}(\delta t_l)\right)^2 \tag{10}$$

where M is either the power-law decay model, the exponential model A, or the exponential model B. $\bar{C}_{ij}(\delta t_l)$ is the average InSAR coherence of a subset of interferograms with the same temporal baseline δt_l. $C_{M,ij}(\delta t_l)$ is the model M predicted InSAR coherence for the temporal baseline δt_l. L is the number of unique δt_l values among all interferograms. We found that the power-law decay and the exponential model B achieved better fitting accuracy (lower MSE) than the exponential model A (see Table V). In addition, the exponential model B leads to noisier temporal decorrelation rate estimates than the other two models [e.g., Fig. 10(a)–(d)].

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It is because the exponential model B has more model parameters than the other two models, which may lead to overfitting due to the presence of other decorrelation factors, such as spatial decorrelation.

We used the InSAR temporal decorrelation rate derived from the three models (α in the power-law decay model and $1/\tau$ in the exponential models) as input features, and we found that the corresponding classification results yield similar surface roughness estimates (see Table V). For example, the ALOS-derived Manning's n maps characterized by different decorrelation models [see Fig. 10(f)–(h)] are consistent with the one derived from the C-CAP 2010 data that are shown in Fig. 10(e). This is because the AdaBoost classification algorithm can combine multiple weak classifiers (e.g., InSAR temporal decorrelation) to produce an accurate strong classifier [36]. As a result, the uncertainties in the InSAR temporal decorrelation rate only have a minor impact on the final classification results. We conclude that the choice of temporal decorrelation models does not introduce a substantial impact on the surface roughness estimation accuracy. In our study, we employed the power-law decay model to estimate the InSAR temporal decorrelation rate because of the superior computational efficiency, as shown in Table V.

D. Performance of Different Classification Algorithms

In both level-I and level-II classifications, the classification is performed at each radar pixel location using a decision-tree-based AdaBoost algorithm. For each training pixel with a known land-cover type, PolSAR and InSAR measurements (see Table III) are used as its feature vector to train the AdaBoost classifiers. The feature vector of each unlabeled pixel is then fed to the classifiers to determine its land-cover class at each level of classification.

We also tested the performance of other classification algorithms, such as random forest and SVM using the ALOS Houston test site [see Fig. 3(a)]. Using each of these two classification algorithms, we trained level-I and level-II classifiers based on the same training pixels [see Fig. 3(b)] and feature vectors (see Table III). We then derived the nine-class land-cover classification results following the two-level classification framework. AdaBoost, random forest, and SVM classification algorithms all result in very similar surface roughness estimates (see Fig. 11). Pearson's correlation coefficients between the C-CAP 2010 surface roughness estimates, AdaBoost, random forest, and SVM estimates are 0.902, 0.901, and 0.900, respectively. We conclude that the performance of the classification framework mostly depends on the input features derived from radar measurements that can discriminate between target land-cover classes, rather than the choice of classification algorithms.

VI. Conclusion

In this study, we developed a new and scalable radar-based land-cover classification algorithm for the application of storm surge modeling, which allows us to map surface roughness along the coast or in a watershed. We note that this algorithm requires multiple repeat-pass radar acquisitions over the same region to characterize the statistical properties of the PolSAR amplitude, coherence, and InSAR temporal decorrelation rate measurements. Given that the ALOS PALSAR satellite has a limited temporal sampling rate (46-day repeat cycle with some data gaps), we could only derive an average surface roughness map over the area of interest around 2010. We were unable to distinguish the land cover and the associated surface roughness changes that occurred during the study period. Scheduled for launch in 2022, the L-band NASA-ISRO Synthetic Aperture Radar (NISAR) mission will provide nearly global PolSAR and InSAR data coverage on a 12-day repeat cycle. The greatly improved temporal coverage of the next generation of L-band spaceborne radar data can be potentially used to derive annual or biannual surface roughness maps, capture land-cover changes in regions undergoing rapid urban development (e.g., the Woodlands and Kingwood areas in North Houston), and inform flooding risks associated with these land-cover changes.

Appendix I
Linearization of the Power-Law Decay Model Fitting

In this section, we describe how to linearize the power-law decay model fitting problem [see (5)]. Suppose that we have positive input vector data $\{x_1, \ldots, x_L\}$ and the observed positive output vector data $\{y_1, \ldots, y_L\}$, and we want to find parameters $\hat{a}$ and $\hat{b}$ to fit a power-law decay model in the least-squares sense such that

$$(\hat{a}, \hat{b}) = \arg\min_{a,b} \sum_{i=1}^{L} \left(y_i - b x_i^a\right)^2. \tag{A.1}$$

The right-hand side of (A.1) can be rewritten as

$$\begin{aligned}\sum_{i=1}^{L} \left(y_i - b x_i^a\right)^2 &= \sum_{i=1}^{L} y_i^2 \left(1 - \frac{b x_i^a}{y_i}\right)^2 \\ &= \sum_{i=1}^{L} y_i^2 \left(1 - \exp\left(\log \frac{b x_i^a}{y_i}\right)\right)^2. \end{aligned} \tag{A.2}$$

Let $\epsilon_i = \log(b x_i^a / y_i)$; we can expand (A.2) as

$$\begin{aligned}\sum_{i=1}^{L} \left(y_i - b x_i^a\right)^2 &= \sum_{i=1}^{L} y_i^2 (1 - \exp(\epsilon_i))^2 \\ &= \sum_{i=1}^{L} y_i^2 \left(1 - \left(1 + \epsilon_i + \mathcal{O}\left(\epsilon_i^2\right)\right)\right)^2 \\ &= \sum_{i=1}^{L} y_i^2 \left(\epsilon_i + \mathcal{O}\left(\epsilon_i^2\right)\right)^2. \end{aligned} \tag{A.3}$$

When the parameters a and b do fit the power-law decay model, we should have $y_i \approx b x_i^a$ such that

$$|\epsilon_i| = \left|\log \frac{b x_i^a}{y_i}\right| \ll 1. \tag{A.4}$$

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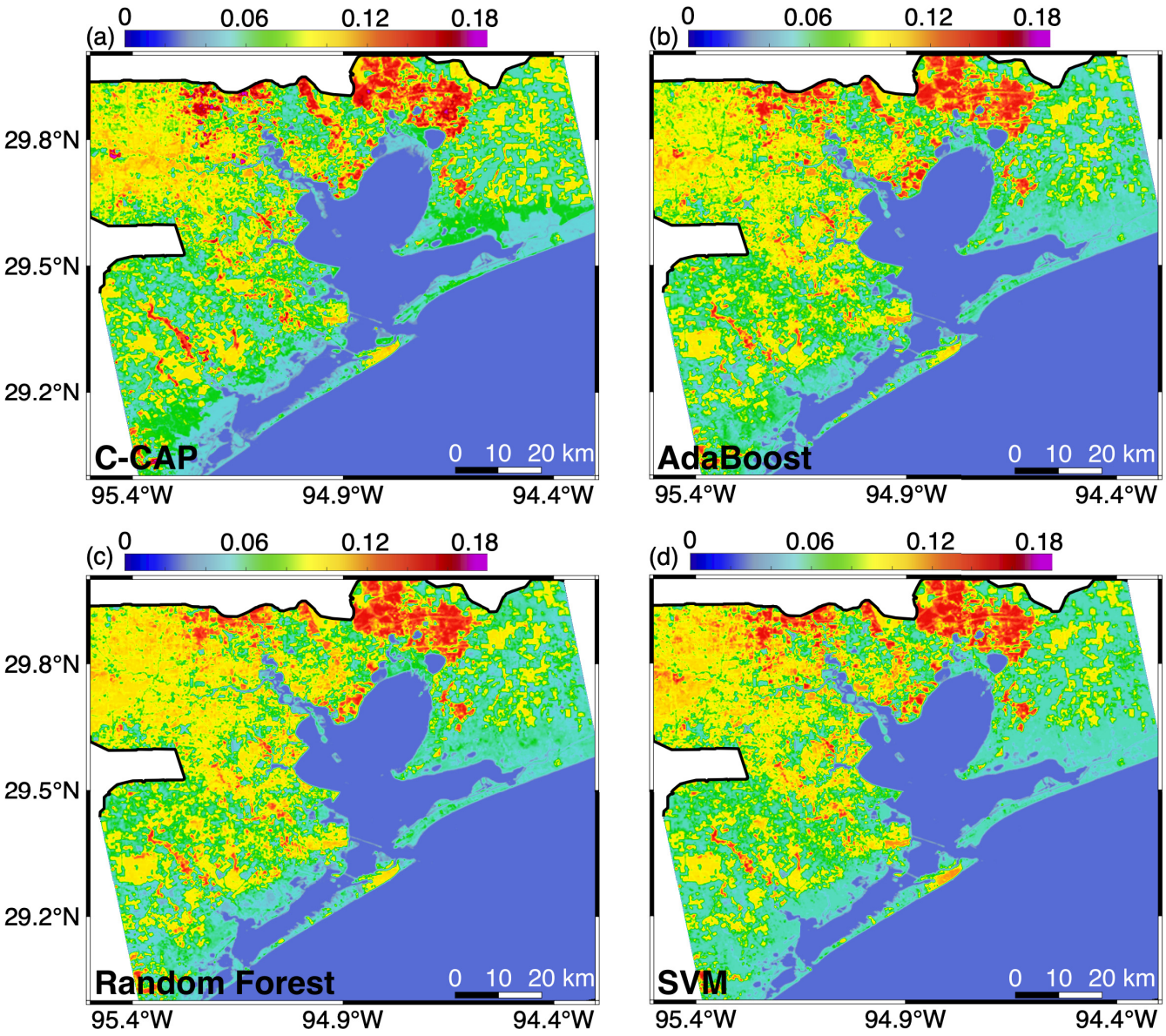

Fig. 11. (a) Manning's n values derived from NOAA's C-CAP 2010 land-cover classification data. Manning's n values derived from ALOS PALSAR data based on (b) AdaBoost, (c) random forest, and (d) SVM classification algorithm. Manning's n values are projected on ADCIRC finite element mesh nodes. The boundary of the computational domain is outlined in black.

Therefore, we can dropout the high order terms of ϵ_i in (A.3), which yields

$$\sum_{i=1}^{L}\left(y_i - bx_i^a\right)^2 \approx \sum_{i=1}^{L} y_i^2\epsilon_i^2 = \sum_{i=1}^{L} y_i^2(\log b + a\log x_i - \log y_i)^2. \quad \text{(A.5)}$$

Substituting (A.5) into (A.1), we finally obtain

$$\left(\hat{a},\hat{b}\right) \approx \arg\min_{a,b} \sum_{i=1}^{L} y_i^2(\log b + a\log x_i - \log y_i)^2. \quad \text{(A.6)}$$

Note that we can introduce a new variable $b' = \log b$ to convert (A.6) into a weighted linear regression problem

$$\left(\hat{a},\hat{b'}\right) \approx \arg\min_{a,b'} \sum_{i=1}^{L} y_i^2\left(b' + a\log x_i - \log y_i\right)^2 \quad \text{(A.7)}$$

where the input data and observed output data are, respectively, $\{\log x_1, \ldots, \log x_L\}$ and $\{\log y_1, \ldots, \log y_L\}$, and the weights for the observations are $\{y_1^2, \ldots, y_L^2\}$. The optimal parameter is $\hat{b'} = \log\hat{b}$. By applying this linearization on the decorrelation rate estimation, (5) can be approximated as

$$(\hat{\alpha}_{ij},\hat{\beta}_{ij}) \approx \arg\min_{\alpha_{ij},\beta_{ij}} \sum_{l=1}^{L} \bar{C}_{ij}(\delta t_l)^2 \left(\log\beta_{ij} - \alpha_{ij}\log\delta t_l - \log\bar{C}_{ij}(\delta t_l)\right)^2 \quad \text{(A.8)}$$

where $\bar{C}_{ij}(\delta t_l)$ is the average InSAR coherence of a subset of interferograms with the same temporal baseline δt_l at the pixel (i, j). L is the number of unique δt_l values. $\hat{\alpha}_{ij}$ is the estimated InSAR temporal decorrelation rate, and $\hat{\beta}_{ij}$ is a scale factor that characterizes the magnitude of the InSAR coherence measurements. After the linearization, the InSAR temporal decorrelation rate at all radar pixels can be calculated

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efficiently using vectorized computation, which can largely reduce the computational cost.

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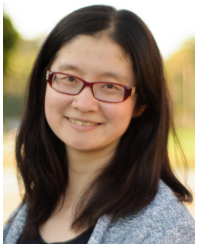

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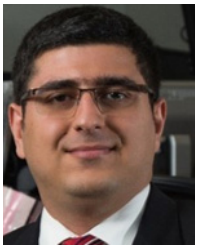

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